# Nibedita Sahu

Data Scientist

Portfolio | GitHub | LinkedIn | Medium

## PROFESSIONAL SUMMARY

Data Scientist focused on building end-to-end data-driven solutions using Python, SQL, statistical modeling, and AI-driven insight generation. I approach problems with structured reasoning, using data to uncover underlying patterns and translate them into actionable decisions. Experienced in data analysis, statistics, and modeling with a strong emphasis on interpretability, business impact, and decision support.

## SKILLS

- **Data Science & Programming:** End-to-end data analysis, feature engineering, and statistical modeling using Python (NumPy, Pandas, Scikit-learn, StatsModels, SciPy) and SQL (MySQL, PostgreSQL).
- **Statistical Modeling & Analytical Techniques:** Exploratory Data Analysis (EDA), probability & statistics, hypothesis testing, regression analysis, and time-series analysis, with a focus on model interpretability, assumptions, and uncertainty handling.
- **Visualization & Decision Communication:** Data visualization and analytical storytelling using Matplotlib, Seaborn, Power BI, and Tableau, along with AI-driven insight generation for business decision support.

## PROJECTS

### Customer Retention & Revenue Optimization System

Python: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn | SQL | Power BI | Generative AI

- Built an end-to-end system to predict customer churn, estimate purchase probability, and optimize targeting strategies under budget constraints using behavior-driven data.
- Combined predictive modeling with decision-focused optimization to identify high-value customers, resulting in significant revenue uplift and measurable business impact.
- Integrated a lightweight AI layer to translate model outputs into business insights, generating actionable recommendations and simulating real-world decision support.

### End-to-End Time Series Analysis & Forecasting with Nutrition Data

Python: NumPy, Pandas, Matplotlib, StatsModels, Scikit-learn | Power BI, Tableau

- Designed a realistic nutrition time-series dataset and performed end-to-end analysis, handling date granularity, trend, and seasonality to understand consumption behavior.
- Applied statistical forecasting models and communicated insights through BI reports and a structured walkthrough, emphasizing interpretability and assumptions.

### Electric Vehicle Population Analysis

Python: NumPy, Pandas, Matplotlib, Seaborn | SQL | Power BI | Canva

- Analyzed large-scale electric vehicle population data using SQL and Python to identify adoption trends, regional distribution, and growth patterns in a policy-focused context.
- Delivered insights through a multi-page Power BI report and presentation, focusing on visualization, storytelling, and clear communication for non-technical stakeholders.

## EDUCATION

**Bachelor of Science in Mathematics**

Kalahandi University, Bhawanipatna | Odisha, India | 2020 - 2023